Sahaj Bhadja

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Summary

Software Engineer with 2+ years investigating production issues, analyzing logs, and fixing defects across **Java**, **C#/.NET**, **Python/Flask**, and **SQL Server** systems on Azure, with Agile Scrum delivery and clear customer-facing communication.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Software Engineer *Los Angeles, CA*

- Investigated and resolved **production failures** in ETL-style integration pipelines across **SQL Server**, PostgreSQL, and Azure storage using reconciliation checks and error handling, sustaining 99%+ reliability across 5K+ monthly jobs.
- Optimized **Azure SQL** queries and indexing for a multi-tenant platform serving 500+ users, cutting dashboard latency from 1.2s to 350ms while isolating async job failures across 1K+ processed AI jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built and supported **Python Flask** REST APIs on AWS serving 150+ instructors and 10K+ daily grading requests, keeping API responses stable during peak load and cutting feedback turnaround 60%.
- Implemented OCR fallback and timeout handling across multimodal LLM providers plus structured per-call **usage logging** in MySQL, giving instructors reliable grading across 10K+ handwritten and text submissions.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- **Troubleshot** large-scale API integrations by **analyzing Jenkins failure logs** and coverage reports from nightly cross-platform runs, then optimizing pagination, filtering, and caching to improve response times 30% for low-bandwidth users.
- Built **object-oriented** Java and Python modules with Singleton and Factory patterns, covered by 20+ Mockito and PyTest unit and integration suites, while collaborating with Microsoft and GitHub stakeholders in **Agile Scrum** sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a resource-management platform with **C# and .NET** RESTful microservices, Azure SQL persistence, and real-time WebSocket updates, synchronizing 2M+ records across locations and reducing resource wastage 42%.
- Diagnosed production errors using **Sentry, Azure Monitor, and Kibana logs**, and shipped fixes through **Azure DevOps** pipelines to Azure App Services with automated unit tests and approval-gated staged releases.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Designed the end-to-end booking architecture, mapped 12+ backend integration points, and **presented it to the client** for alignment, then built **Flask** APIs keeping booking responses under 200ms across 50K+ records.
- Automated refund, dispute, and payment-status handling through **idempotent webhooks**, cutting manual refund work about 70% and preventing double bookings with Firebase transactional locking on shared time slots.

Skills

Languages: Java, C++, C#, Python, SQL, JavaScript, TypeScript
Frameworks & Data: .NET, Flask, Spring Boot, REST APIs, SQL Server, Azure SQL, PostgreSQL, MySQL
Support & Debugging: Log Analysis, Root Cause Analysis, Debugging, Troubleshooting, Sentry, Azure Monitor, Kibana, Customer Communication
Cloud & Networking: Azure, Azure App Services, AWS, Docker, TCP/IP, Linux, Windows
Tools & Practices: Jira, Azure DevOps, Git, Jenkins, Unit Testing, PyTest, Mockito, Code Reviews, Agile/Scrum, Design Patterns

Projects

Log Triage and Defect Reporting Tool | *Python, Flask, SQL Server, Jira REST API, Regex*

- Designed a Flask service that parses application and device logs, groups recurring error signatures by version and site, and drafts **Jira defect tickets** with reproduction steps, **root cause analysis**, and proposed fixes.

Device Network Traffic Analyzer | *C++, Python, TCP/UDP Sockets, pcap, SQL Server*

- Designed a C++ tool capturing **TCP and UDP** traffic between simulated automation stations, flagging dropped, delayed, or out-of-order messages, and storing session timelines in SQL Server for **network traffic analysis**.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*

- Won Smart India Hackathon with a **hospital** management platform integrating ABDM APIs for consent-based patient history, Twilio appointment alerts, and **Sentry monitoring**, reducing administrative workload 30% across 100+ daily appointments.